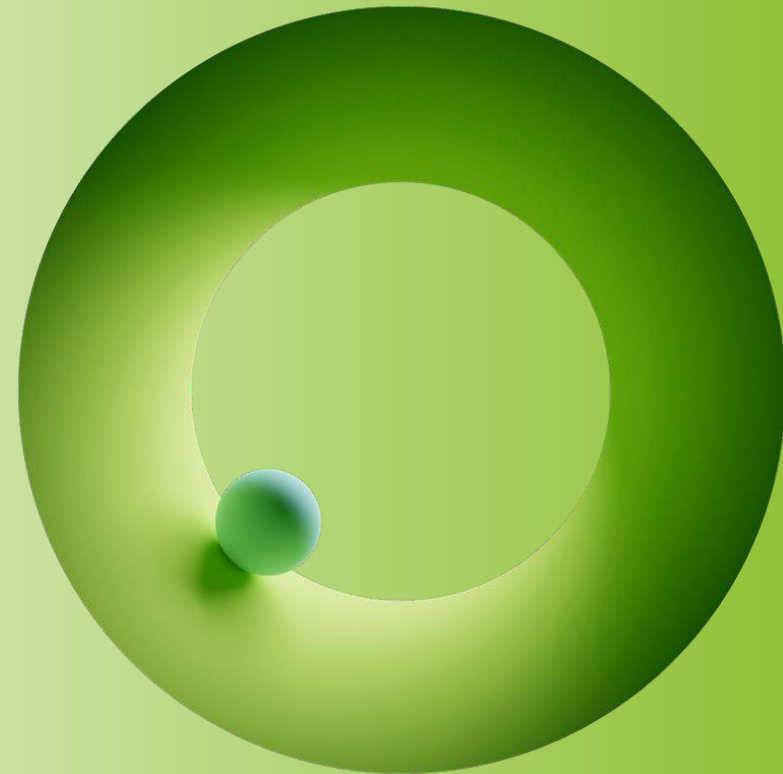
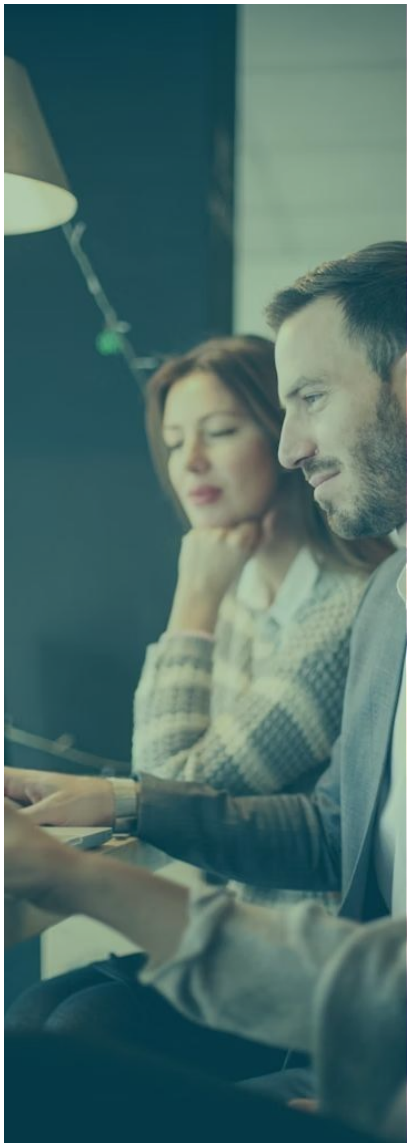




**Use Case Implementation –
TUM Project Study 2026**
Requirements for Agent implementation



Project Study 2026

Deloitte. **TUM**



Steven Liu



Katja Panko



Yan-Ru Wang



Sören Lange-Klapproth

Deloitte.



Marcus Pankow
Partner



Julian David Kaiser
Senior Consultant



Kai Sumida
Consultant



Paul Axmann
Consultant

Implementation: Automated Reporting & Status Updates

Many experts explicitly stated that this is where AI provides the biggest productivity gains.

**Prio #1 for
Implementation**

Examples

- **Interview #2**
 - **reporting is already where they successfully use AI and sees it as an obvious efficiency gain.**
- **Interview #4**
 - **AI should automatically prepare emails, status reports and reminders instead of humans doing administrative work.**
- **Interview #5**
 - **automatic management reports prevent forgotten tasks and improve reporting quality.**
- **Interview #7**
 - **reporting is the biggest opportunity because it is highly data-driven.**
- **Interview #9**
 - **"Getting rid of administrative work" was explicitly identified as the number one AI opportunity.**

Implementation: PMI Project Tracking & Monitoring

Interviewees didn't want AI to manage projects, but wanted AI to reduce the administrative effort of coordinating project execution.

**Prio #2 for
Implementation**

Examples

- **Interview #2**
 - **AI should suggest governance, meeting cadence and project setup instead of starting from a blank page.**
- **Interview #7**
 - **Main responsibility is tracking onboarding, timelines, project progress, AI could automate much of this monitoring.**
- **Interview #4**
 - **AI should monitor milestones and automatically remind responsible people.**
- **Interview #6**
 - **Already implementing AI-supported task creation from meeting transcripts into Jira.**

AI Agent: Automated Reporting & Status Updates

What to implement

1. User uploads PMI files and requests an output
Example:
"Create a SteerCo PowerPoint"
"Create a Finance Excel dashboard"
"Create a chart about risks"
2. Agent asks for role/audience
Executive / PMO / Finance
If already clear from the request, skip this question.
3. Agent extracts data
From Excel, PowerPoint, PDF, Word, HTML.
4. Agent standardizes data
Into one PMI data model:
tasks, milestones, risks, budget, KPIs, owners, dates, status.
5. Agent runs consistency checks
Example:
Excel says progress = 82%
PowerPoint says progress = 75%
6. Agent handles conflicts
Option A: ask user to resolve critical conflicts.
Option B: apply predefined source priority:
Excel > Word/PDF > PowerPoint > HTML
7. Agent generates the requested output
Depending on user request:
 - PowerPoint presentation
 - Excel dashboard
 - Specific chart

AI Agent: Automated Reporting & Status Updates

How to implement

Programming Language	Python	Word Processing	python-docx	Database (Optional)	SQLite
Backend Framework	FastAPI	PDF Processing	PyMuPDF (fitz) pdfplumber	Logging	Python logging
Alternative Frontend	React + TypeScript	HTML Processing	BeautifulSoup4	Deployment (Production)	Docker
LLM	OpenAI GPT-5.5 (API)	Chart Generation	matplotlib Plotly (optional)		
Agent Framework	LangGraph	Image Handling	Pillow		
Structured Output	Pydantic	Data Validation	Pydantic		
Excel Processing	pandas openpyxl XlsxWriter	Workflow Logic	Python classes/functions		
PowerPoint Processing	python-pptx	Storage (Prototype)	Local JSON		

Costs: all free

only OpenAI API
20 EUR/ month

What will you receive

At the end of the project you will receive the following

1. **Frontend App** - or we can make all in console like in claude code project
2. **GitHub Repository**
3. **Docker**
4. **User Guide**
5. **Sample Data**
6. **Example Outputs**
7. A **requirements.txt** file for Python dependencies.
8. A **README** explaining how to start the application and use it.